

BRONNOY, Norway

WMO: 011120

Lat: 65.457N

Lon: 12.218E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.8	-6.2	-15.5	1.0	-5.7	-13.7	1.1	-4.9	17.4	0.6	15.6	0.2	3.4	40	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	24.1	16.6	21.9	15.4	19.9	14.5	17.5	22.8	16.0	20.4	15.0	18.8	4.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.8	19.4	14.1	10.1	17.5	13.2	9.5	16.4	49.2	22.9	44.8	20.4	41.9	19.0	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
13.4	11.6	10.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	-10.6	19.1	1.4	2.0	-11.6	20.6	-12.4	21.7	-13.2	22.9	-14.2	24.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	0.3	0.8	1.5	4.4	8.3	10.9	13.9	13.7	11.5	7.2	3.5	1.7
	DBStd	5.86	3.63	3.21	3.00	3.13	3.71	2.97	3.54	2.93	2.90	2.98	3.30	3.23
	HDD10.0	1657	302	256	262	171	78	20	2	1	16	96	196	258
	HDD18.3	4328	560	489	521	417	310	223	148	147	205	346	445	516
	CDD10.0	387	0	0	0	4	27	48	122	116	62	8	1	0
	CDD18.3	17	0	0	0	0	1	1	11	4	1	0	0	0
	CDH23.3	103	0	0	0	0	3	3	83	14	0	0	0	0
	CDH26.7	15	0	0	0	0	0	0	15	0	0	0	0	0
Wind	WSAvg	4.6	5.4	5.5	5.2	4.4	4.2	4.0	3.5	3.7	4.1	4.8	5.1	5.5
Precipitation	PrecAvg	1409	134	121	128	88	70	91	99	105	151	161	122	151
	PrecMax	1759	251	272	271	160	142	174	216	195	372	299	245	266
	PrecMin	1098	0	1	45	22	25	28	47	29	44	40	20	8
	PrecStd	198	64	65	60	42	26	42	46	40	82	62	72	70
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	8.1	9.1	17.5	22.4	23.9	28.4	25.1	20.8	16.3	12.4	8.6
		MCWB	5.0	5.1	4.6	9.7	14.6	14.2	19.9	16.7	14.2	11.3	8.9	5.5
	2%	DB	7.0	6.9	7.2	13.3	19.7	21.0	25.7	22.9	19.0	14.3	10.1	7.7
		MCWB	5.4	5.2	4.7	7.6	12.6	13.9	17.5	15.9	13.9	10.1	7.6	6.1
	5%	DB	6.1	6.1	6.3	10.9	17.0	18.1	22.8	20.9	17.7	12.9	9.1	7.1
		MCWB	4.6	4.5	4.7	6.5	11.5	12.9	16.9	15.1	13.0	9.5	6.8	5.6
	10%	DB	5.2	5.1	5.8	9.1	14.2	16.1	20.0	19.0	16.0	11.3	8.1	6.2
		MCWB	3.6	3.6	4.3	5.4	9.8	11.8	15.5	14.3	12.3	8.6	6.0	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	6.6	6.6	10.0	15.3	16.0	20.6	18.1	15.6	12.1	9.9	7.7
		MCDB	7.0	7.0	6.8	16.8	22.0	21.3	27.0	23.5	19.3	15.3	10.7	7.8
	2%	WB	5.8	5.6	5.8	8.2	13.1	14.4	18.6	16.5	14.3	10.9	8.3	6.9
		MCDB	6.3	6.3	6.3	11.9	18.7	19.3	24.4	21.4	17.8	13.6	9.7	7.2
	5%	WB	4.9	4.8	5.1	7.0	11.7	13.3	17.1	15.4	13.5	10.0	7.3	5.8
		MCDB	5.7	5.7	6.1	9.6	16.3	17.4	21.7	19.7	16.6	12.0	8.7	6.5
	10%	WB	3.7	3.7	4.2	6.0	10.2	12.2	15.6	14.6	12.7	9.1	6.1	4.5
		MCDB	5.0	4.8	5.5	8.1	13.8	15.4	19.2	18.1	15.3	11.0	7.8	5.9
Mean Daily Temperature Range	5% DB	MDBR	3.7	3.8	4.4	5.3	6.1	5.5	6.1	5.8	5.5	4.4	3.6	3.6
		MCDBR	4.0	4.3	4.3	8.0	10.4	9.9	11.8	9.2	7.6	5.9	4.3	4.3
		MCWBR	3.5	3.9	3.6	4.5	5.0	4.9	5.3	4.6	4.0	3.6	3.3	4.0
	5% WB	MCDBR	3.9	4.2	3.9	6.6	9.7	8.5	10.8	8.2	6.7	5.0	4.3	4.2
		MCWBR	3.9	4.2	3.8	4.2	5.2	4.6	5.2	4.4	3.9	3.8	3.8	4.3
Clear-Sky Solar Irradiance	taub	0.367	0.296	0.324	0.348	0.348	0.347	0.355	0.342	0.330	0.305	0.346	0.364	
	taud	2.570	2.203	2.282	2.331	2.429	2.494	2.499	2.523	2.535	2.369	2.535	2.527	
	Ebn at Noon	251	574	743	818	858	867	848	825	753	573	242	72	
	Edh at Noon	22	54	80	96	96	93	90	80	64	46	20	10	
All-Sky Solar Radiation	RadAvg	0.08	0.47	1.73	3.44	4.56	4.84	4.52	3.53	1.88	0.76	0.16	0.02	
	RadStd	0.01	0.09	0.27	0.37	0.32	0.59	0.54	0.38	0.26	0.13	0.02	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page